

**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 5

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Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission

- To impart quality education in Computer Science and Engineering by means of learning techniques and value-add courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interactions.
- To build leadership skills, ethical values and teamwork among the students.
- To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

1) Internet Access Architecture

- * Internet Access Services are provided by ISP while backbones are provided by NSPs.
- * It is defined as a meta network, a constantly changing collection of thousands of individual network intercommunicating with a common protocol.
- * The Architecture is based in the very specification of the standard TCP/IP protocol designed to connect any two networks which may be very different in internal hardware, software and Technical design.
- * Internet communication in the united state is achieved by connecting the backbone at four major NAPs. The companies involved are

- Amerite - Chicago
- Pacific Bell - San Francisco
- MFS Datnet - Washington D.C
- Sprint - Pennsauken, New Jersey

An ISP must either connect directly to an NSP

Routing Arbiters (RAs)

The RA is the organization that provide routing information at each NAP. The RA provides customised routing information at each NAP which reflects all bilateral arguments between the NAP's clients.

Route Servers

The route servers exchange routing information with the ISP routers attached to NAP. customized routing views implement the individual routing policy requirements of each ISP.

Network Management System

Distributed routers run on each of the route servers and collect performance statistics, delay matrix measurements, and throughput measurements.

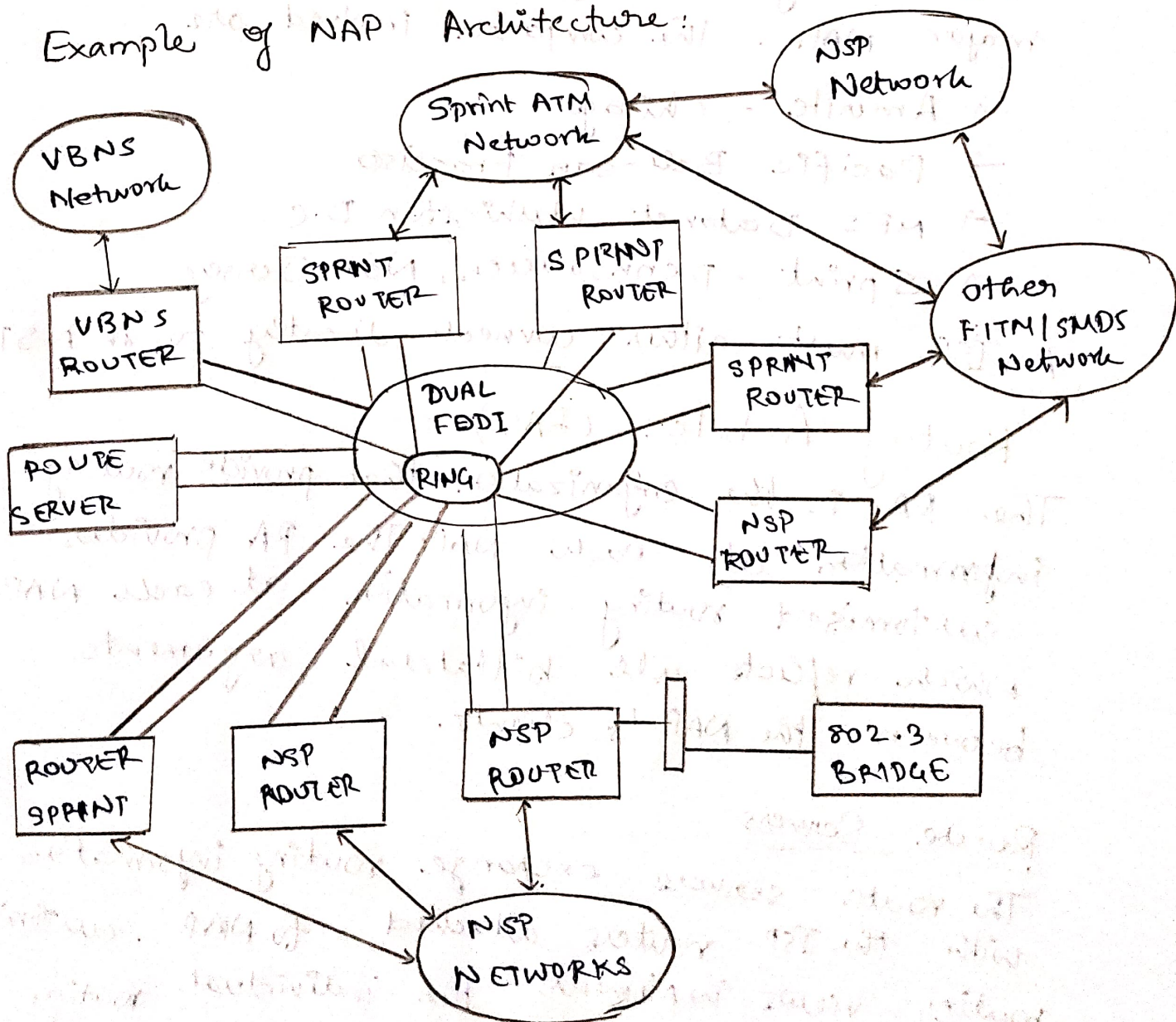
Routing Arbiter database

One of several routing databases collectively known as the Internet Routing Registry.

Routing engineering:

The RA team works with NSP/ISPs and IT providers to set up peering and resolve routing problems at the NAP's.

Example of NAP Architecture:



NAP Access

Sprint supports access from an NSP's router located at a sprint's physical NAP facility. The NSP can supply and maintain the router which interfaces directly with sprint's NAS, or it can choose a sprint-supplied router.

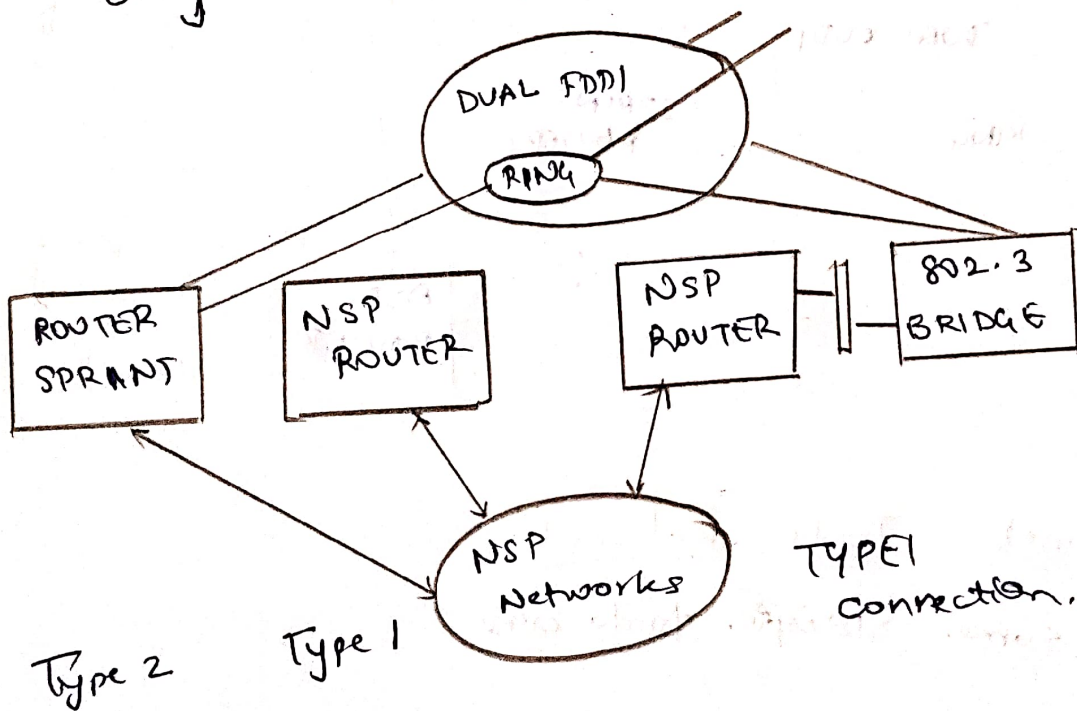
The WAN technologies supported includes.

* Dedicated circuit Access

* Switched service Access.

Dedicated Circuit Access

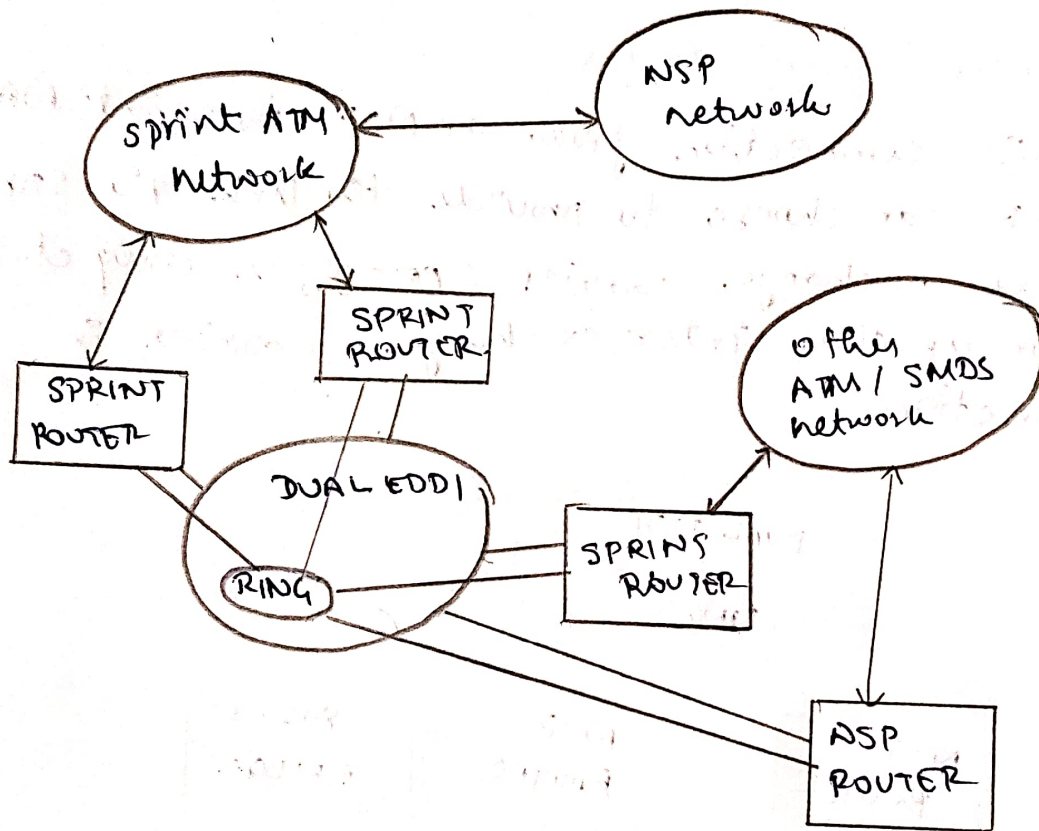
For this connection from an NSP to sprint NAP, then NSP can choose to provide the link via sprint or local exchange carrier (LEC), a competitive LEC or another inter exchange carrier in conjunction.



Switched service Access:

The Sprint NAP may also be accessed by any one of three switched services. ATM, frame relay, or SMDS.

The NSP connects directly via an SMDS line or network to Sprint - provided router. An NSP may use B-ICI to connect another carrier's frame relay on Sprint ATM.



2) Internet Tools Relevant

The some specific tools are.

* Yahoo!

This is currently considered the most complete directory resource on the world wide web. It is a categorized, menu-driven directory of web resources. users can also execute an automatic search on their databases from their main pages.

General search engines on the web are discussed their databases are created by automatic robots which travel the web on regular basis.

* Infoseek:

This search engine accepts natural language questions such as, "What is good restaurant in New York city that serve fresh meals!" but also provides keyword search capability.

* Altavista:

This is a search engine that is very linkable often it may be all a user needs.

* Open Text: This is a technically superior search engine on Internet and produces fast, relevant responses.

* Web crawler:

Web crawler is faster than Infoseek and lets the user view more matches on screen simultaneously. It does not screen its matches for relevance as much as Infoseek does.

* Lycos:

Lycos is the slowest of the big three, but it is also the most detailed. The Lycos robot indexes not only the sites that it visits, but also every link on those sites.

* Sury Search

One can search all the previously mentioned databases simultaneously using a tool called surysearch.

* Galaxy

Galaxy searches for resources on the web, including Gopher and Telenet.

* Vsenet

You can read news directly or search a vsenet Archive such as Deja News for what you are looking for.

* Archie:

If you are searching for computer software, then Archie is an excellent resource for navigating through the content on anonymous FTP archives throughout the world.

* Gopher:

Gopher is a service that allows users to view and obtain files, programs and software.

* Veronica:

Veronica is a self-updating database of Gopher documents. It allows users to search all the gopher sites.

* FTP:

Anonymous FTP permits users to access remote systems without actually having user accounts on the system. Users can FTP to any anonymous sites in the world using the

hypertext Interface.

Http : // hoo.hoo.nesa.uiuc.edu: 80 /ftp - Interface, html.

* Tele net :

Tele net Allows users to remotely log in to other computers and can give them access to databases and many information services. For example, users can search the full ERIC database from 1966 at the 3270:11 andee cad. duc.aurun.edu

* WAIS

The WAIS lets the user search indexed info to find articles containing groups of chosen words. WAIS is like Gopher in that it shields users from having to know on computer the information resides but unlike Gopher.

3) Searching the Internet

As the amount of content on-line increases, it becomes more and more difficult for users to find what they are looking for on the internet.

Search Engine :

The term search engine is used to define the entire search service and process on a database provided by a particular vendor.

Actually, the search engine is just a part of the process. The process consists of three main components.

- * Sourcing Information.

- * Index, catalog or database creation.

- * The search Engine.

Spider

A spider also known as a robot or crawler is actually just a program that follows or crawls link throughout the internet, grabbing content from sites and adding it search engine indexes.

- + Internet search processes fall into two groups based upon how the database or index is compiled: meta index or spider generated index. Spider generated index systems provide much the same services as metaindex system but does so through a program called a spider.

Uses of spiders

The primary uses of spiders are:

- + Link validation.
- + HTML validation.
- + 'What's new' monitoring.
- Indexing.

Problems:

- + The web is that references to other pages may become dead links when the referenced page is moved or removed.
- + There is no formal process to notify web pages that have referral links that those links may have changed.
- + A spider can also check to make sure documents are in proper HTML format before submitting the link back to some data catalog.

Applications

- * It concerns resource discovery
- * There are several search processes that use spiders to summarize large parts of internet.
- * Lycos and Inposeek are two of most well known spider-based processes.

Advantages

- * Databases can be updated automatically, allowing users to be reasonably sure they are receiving the latest information.
- * This is a contrast to manual maintenance effort where a full updating may take place only once or twice a year and may not completely cover the entire database.

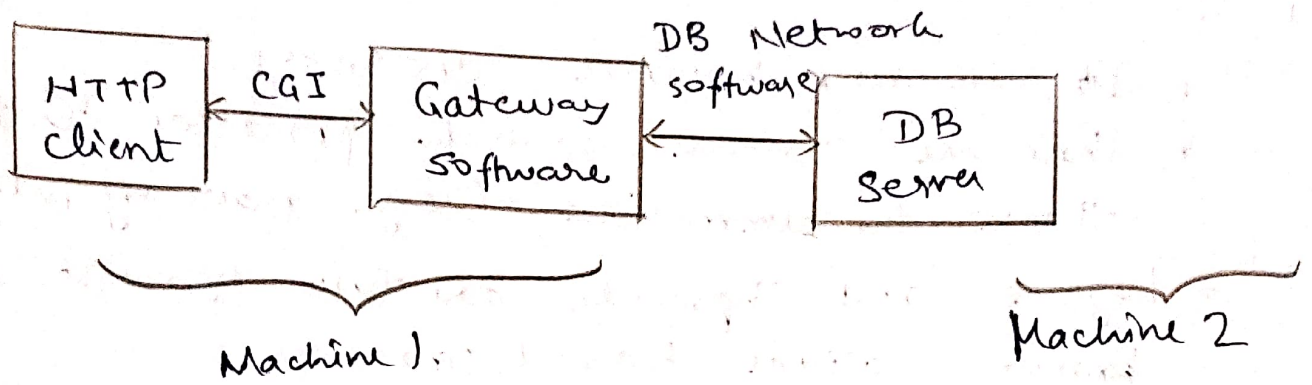
Limitations

- * There is just too much information available through the internet and amount is growing all the time.

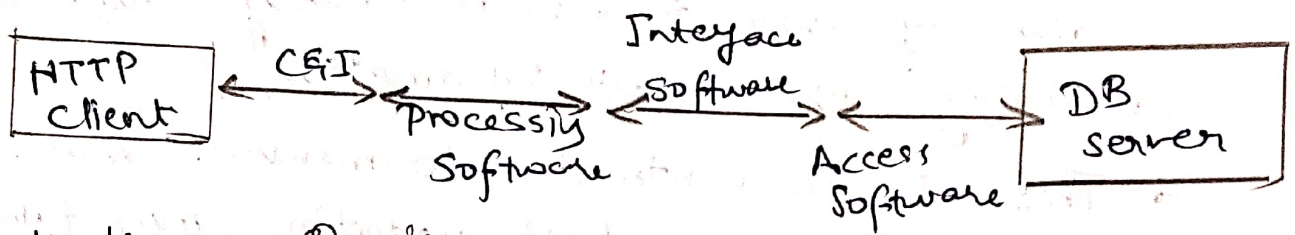
Although spider generated databases are generally more current in totality than manually compiled databases, the fact that new information is being added every day precludes the spider generated database from being completely up-to-date.

Accessing the Information Databases:

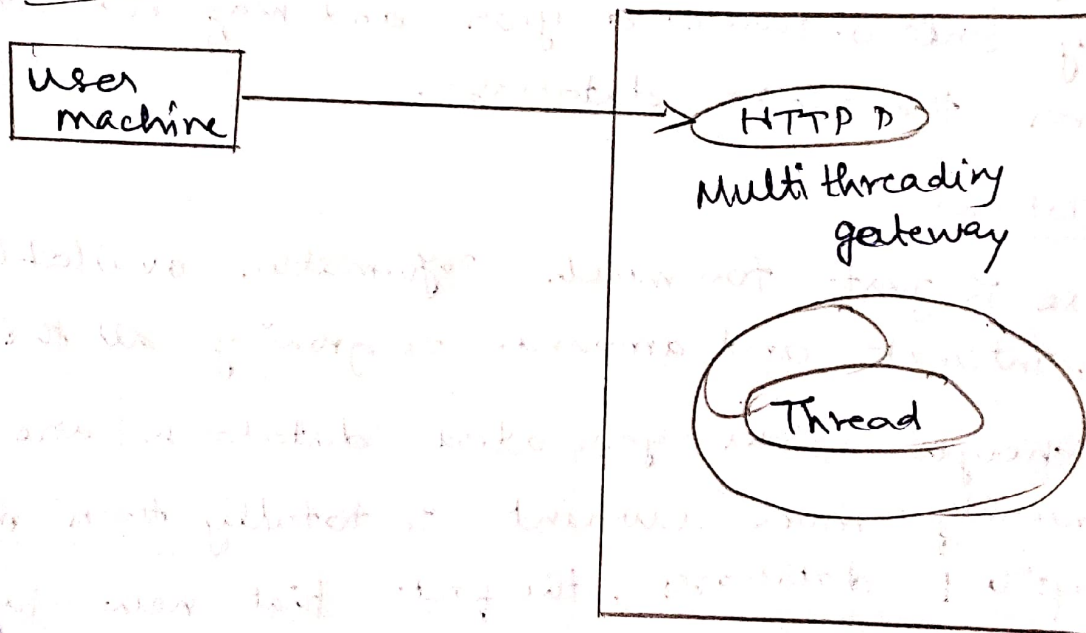
An HTTP server provides a method of running database searches from within an HTML document.



CGI Use:



Multi thread Queries:



Multi threading Queries are performed by using a gateway program which conforms to CGI.

The program works with an HTML form.

Features :

- * Free-Text-Search
- * Automatic morphology
- * Word indexing
- * Lexical affinity.
- * Ranking and relevance scoring.

The some common indexing systems are,

- * ALIWEB (Archie Like Indexing in the web)
- * SWISH (Simple Web Indexing for Humans)
- * WAIS
- * Archie.